

Proposition 2 for the Reference Spec

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This note answers two questions for the reference firm regression

`firm_main__levels__extensive__coalition__cycle_specific__unweighted__all_firms__pooled_coun`

1. How can we show Proposition 2 really holds in the data?
2. Why does the simple regression on cell means still give different coefficients?

All numbers below come from:

- `BNDES/output/diagnostics/prop2_reference_spec_comparison.csv`
- `BNDES/output/diagnostics/prop2_reference_spec_diagnostics.csv`
- `BNDES/output/diagnostics/prop2_reference_spec_cell_fe.qs2`

Short Answer

Proposition 2 **does hold** if we start from the firm regression itself, keep the exact estimation sample that the firm regression used, save the fitted fixed effects, and aggregate the **firm sufficient statistics**.

Proposition 2 **does not hold** if we replace that object with a simpler new regression on cell means. That simpler regression loses important within-cell variation.

So there are really two different exercises:

- **Exact aggregation test:** this is the right test of Proposition 2. It now works.
- **Simple cell-mean regression:** this is only an approximation. On real data it still differs.

1 The Two Samples: Group A and Group B

Define

$$A = \{i : y_i \text{ and } x_i \text{ are observed}\},$$
$$B = \{i \in A : i \text{ is actually used by the firm regression}\}.$$

Here:

- $|A| = 44,181,405$
- $|B| = 43,184,682$
- $|A \setminus B| = 996,723$

So the firm regression does **not** use all rows in A . It uses only B .

This matters because the aggregate version must be built from B , not from A . If we aggregate A but the firm regression used B , then the two sides are not even using the same data.

The new script now fixes this directly:

- fully kept firms: 5,464,381
- partially dropped firms: 121
- fully dropped firms: 996,453
- fully dropped singleton firms: 996,450

In simple language: almost all dropped firms disappear completely from the firm regression, so they should also disappear from the aggregated version.

2 The Exact Firm Equation

On the used sample B , the firm regression is

$$y_i = \beta x_i + FE_i + u_i,$$

where:

- $y_i = \mathbf{1}(\text{BNDES}_{fmt} > 0)$
- x_i is one of FA_mayor_coalition, FA_gov_coalition, FA_pres_coalition
- FE_i is the fitted sum of the firm fixed effect and the municipality-year fixed effect

After estimating the firm regression, define

$$y_i^* = y_i - \widehat{FE}_i.$$

Then, with one regressor, the firm coefficient is

$$\hat{\beta}_{\text{firm}} = \frac{\sum_{i \in B} x_i y_i^*}{\sum_{i \in B} x_i^2}.$$

This is the key formula.

3 How to Show Proposition 2 Holds Exactly

Partition the used sample B into cells $c = (j, m, t)$. For each cell define

$$S_{xy,c} = \sum_{i \in B_c} x_i y_i^*, \quad S_{xx,c} = \sum_{i \in B_c} x_i^2.$$

Then

$$\hat{\beta}_{\text{firm}} = \frac{\sum_c S_{xy,c}}{\sum_c S_{xx,c}}.$$

This is just regrouping the same firm sums by cell. Nothing is approximated.

This is the exact way to show Proposition 2 holds in code:

1. Run the firm regression.
2. Save the exact used sample B .
3. Save the fitted fixed effects $\widehat{FE}_i$.
4. Compute $y_i^* = y_i - \widehat{FE}_i$.
5. Aggregate $S_{xy,c}$ and $S_{xx,c}$ by cell.
6. Rebuild $\hat{\beta}$ from those cell sums.

That is exactly what the new script does.

Result

Tier	Firm beta	Exact beta from cell sums	Exact gap
Mayor	0.002176	0.002176	-2.20e-11
Governor	0.003404	0.003404	-6.56e-12
President	0.000378	0.000378	1.86e-12

The largest cell-level identity error is only 1.11e-16.

So this part is now clear:

If we carry the exact firm regression solution to the cell level, Proposition 2 holds for the reference spec.

4 Why the Simple Regression on Cell Means Still Fails

Now define cell means on the used sample B :

$$\bar{x}_c = \frac{1}{N_c} \sum_{i \in B_c} x_i, \quad \bar{y}_c^* = \frac{1}{N_c} \sum_{i \in B_c} y_i^*.$$

The simple aggregated regression uses

$$\hat{\beta}_{\text{mean}} = \frac{\sum_c N_c \bar{x}_c \bar{y}_c^*}{\sum_c N_c \bar{x}_c^2}.$$

This looks similar, but it is **not** the same object as the exact firm formula.

The reason is:

$$\sum_{i \in B_c} x_i^2 = N_c \bar{x}_c^2 + \sum_{i \in B_c} (x_i - \bar{x}_c)^2,$$

and

$$\sum_{i \in B_c} x_i y_i^* = N_c \bar{x}_c \bar{y}_c^* + \sum_{i \in B_c} (x_i - \bar{x}_c)(y_i^* - \bar{y}_c^*).$$

So the simple mean regression drops two within-cell terms:

- within-cell variance of x
- within-cell covariance between x and y^*

If those within-cell terms are small, the mean regression is close. If those terms are large, the mean regression is different.

In this dataset they are large.

Direct Evidence

The share of the exact denominator that comes from the dropped within-cell part is:

- Mayor: 91.9%
- Governor: 91.9%
- President: 93.7%

This means most of the identifying variation is **inside** cells, not between cell means.
That is why the simple mean regression still differs:

Tier	Firm beta	Mean-regression beta	Gap
Mayor	0.002176	0.001290	-0.000886
Governor	0.003404	0.004397	0.000993
President	0.000378	-0.006601	-0.006979

So the mean regression is not failing because Proposition 2 is false. It is failing because cell means are not enough to summarize the firm-level problem.

5 C1 to C6 in Simple Language

Below is the clean way to read the conditions from the plan.

C1. Correct weighting

Plain language: a cell with 100 firms should count 100 times more than a cell with 1 firm.

In the mean regression this means using weight `N_c`.

Status here: **enforced**.

C2. Same sample

Plain language: the aggregated version must use the exact same firm rows as the firm regression.

Here that means: aggregate *B*, not *A*.

Status here: **enforced**.

C3. Firm immobility / FE nesting

Plain language: if the same firm appears in more than one cell, then one simple cell fixed effect cannot play the role of that firm's fixed effect.

Status here:

- **not needed for the exact aggregation test**, because we keep the firm FE from the firm regression;
- **still a problem for a separate simple cell FE regression**.

C4. Correct aggregated FE object

Plain language: the aggregated nuisance term must be the one implied by the firm regression, not a guessed simpler fixed effect.

Status here: **enforced in the exact test**, because we use the saved fitted FE from the firm regression and average it to the cell level.

C5. Fixed cell composition

Plain language: if the firms inside a cell change over time, then the average firm fixed effect of that cell also changes over time.

Status here:

- **not needed for the exact aggregation test**, because we work cell-year by cell-year using the saved firm solution;
- **still a problem for a separate simple cell FE regression**.

C6. No within-cell regressor heterogeneity after FE removal

Plain language: once fixed effects are removed, firms inside the same cell should all have the same x if cell means are supposed to be enough.

Status here: **fails on real data**, and this is the main reason the simple mean regression is still different.

6 What Should Be Shown in the Paper or Presentation

The clearest presentation is:

1. Show the firm regression.
2. Define A and B , and state that the aggregation uses B .
3. Show the exact identity

$$\hat{\beta} = \frac{\sum_c S_{xy,c}}{\sum_c S_{xx,c}}.$$

4. Report that the exact cell-sum coefficient matches the firm coefficient up to machine precision.
5. Then show the mean-regression coefficient separately and explain that it differs because the dropped within-cell terms are large.

That tells the full story very simply:

Proposition 2 holds exactly when we aggregate the firm regression itself. A new regression on cell means is only an approximation, and on real data that approximation is not exact because within-cell variation is large.